

Chapter Three

Crystal Binding

3.2 Ionic Crystals and Electrostatic or Madelung Energy

Ionic crystals are made up of positive and negative ions. The **ionic bond** results from the electrostatic interaction of oppositely charged ions. Two common crystal structure structures found in ionic crystals, the sodium chloride and the cesium chloride.

Madelung energy The long range interaction between ions with $\pm q$ is the electrostatic interaction $\pm \frac{q^2}{r}$, attractive between ions with opposite charge and repulsive between ions of same charge. The repulsive interaction between ions are same as inert gas atom but Vander Waal part of interaction has relatively small contribution to the cohesive interaction in ionic crystal is order of 1 or 2 percent. The main contribution to the binding energy of ionic crystals is electrostatic and is called the Madelung energy.

If U_{ij} is the interaction energy between ions i and j , we define a sum U_i which includes all interaction involving the ion i :

$$U_i = \sum_j U_{ij}, \text{ where the summation includes all ions except } j = i.$$

we suppose that U_{ij} may be written as sum of a central field repulsive potential of form $\lambda \exp(-r/\rho)$, where λ and ρ are empirical parameters, and a coulomb potential $\pm \frac{q^2}{r}$.

Thus, in CGS unit $U_{ij} = \lambda \exp\left(-\frac{r_{ij}}{\rho}\right) \pm \frac{q^2}{r_{ij}}$, where the positive sign is taken for like charges and negative sign is for unlike charges. In SI units the coulomb interaction is $\pm \frac{q^2}{4\pi\epsilon_0 r_{ij}}$.

It is convenient to introduce quantities p_{ij} such that $r_{ij} = p_{ij}R$ where R is the nearest-neighbour separation in the crystal. If we include repulsive interaction only among nearest neighbours, we have

$$U_{ij} = \begin{cases} \lambda \exp\left(-\frac{R}{\rho}\right) - \frac{q^2}{R} & (\text{nearest neighbors}) \\ \pm \frac{1}{p_{ij}} \frac{q^2}{R} & (\text{otherwise}) \end{cases}$$

$U_{total} = NU_i = N\left(z\lambda \exp\left(-\frac{R}{\rho}\right) - \frac{\alpha q^2}{R}\right)$, where z is the number of nearest neighbours of any ion and $\alpha = \sum_j \frac{(\pm)}{p_{ij}}$ is identified as Madelung constant.

For simplicity we are taking $U = N\left[-\frac{\alpha q^2}{4\pi\epsilon_0 R} + B \exp\left(-\frac{R}{\rho}\right)\right]$

Madlung energy

$$U_{total} = NU_i = N\left(z\lambda \exp\left(-\frac{R}{\rho}\right) - \frac{\alpha q^2}{R}\right)$$

$$N \frac{dU_i}{dR} = -\frac{Nz\lambda}{\rho} \exp\left(-\frac{R}{\rho}\right) + \frac{N\alpha q^2}{R^2} = 0$$

$$R_0^2 \exp\left(-\frac{R_0}{\rho}\right) = \rho \alpha \frac{q^2}{z\lambda} \text{ to determine equilibrium separation } R_0 \text{ if the parameters } \rho, \lambda \text{ of}$$

repulsive interaction are known. For SI unit replace q^2 by $\frac{q^2}{4\pi\epsilon_0}$.

The total lattice energy of crystal $2N$ ions at their equilibrium separation R_0 may be written as

$$U_{tot} = -\frac{N\alpha q^2}{R_0} \left(1 - \frac{\rho}{R_0}\right). \text{ The term } -\frac{N\alpha q^2}{R_0} \text{ is identified as Madlung energy. We can find that}$$

ρ is of order of $0.1R_0$. so that repulsive interaction has very short range. In some place

Madelung constant is also represented A . In SI unit $U_{tot} = -\frac{N\alpha q^2}{4\pi\epsilon_0 R_0} \left(1 - \frac{\rho}{R_0}\right) kJ / kmole$. Where

N is Avogadro number

Evaluation of the Madelung constant

The Madelung constant is given by $\alpha = \sum_j \frac{(\pm)}{p_{ij}}$ if we take reference ion as a negative charge the

plus sign will apply to positive ions and minus sign to negative ions the equivalent definition is

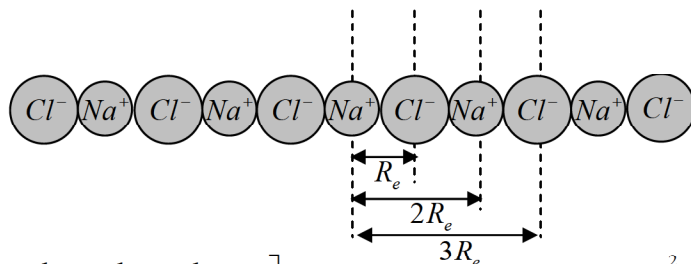
$\frac{\alpha}{R} = \sum_j \frac{\pm}{r_j}$ where r_j is j^{th} ions from the reference ion and R is the nearest neighbour distance

It must be emphasized that the value given for α will depend on whether it is defined in term of the largest neighbour distance R or in terms of the lattice parameter a or in term of relevant length.

Example: Show that the Madelung constant α for an infinite linear chain of ions of alternating charge at an equilibrium separation is $2 \ln 2$.

Solution: Madelung constant A is a function of crystal structure and can therefore be calculated from the geometrical arrangement of ions in a crystal. In the present case, we have to calculate the Madelung constant for an infinite linear chain of ions of alternating charge (i.e. a one dimensional $NaCl$ crystal) at equilibrium separation R_0 . Let us take one sodium ion as the reference ion and then write various terms of the series for alternating ions in terms of R_0 ,

we obtain



$$\frac{\alpha}{R_0} = 2 \left[\frac{1}{R_0} - \frac{1}{2R_0} + \frac{1}{3R_0} - \frac{1}{4R_0} + \dots \right] \quad \text{We know that } \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$$\Rightarrow \alpha = 2 \left[1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \right] \Rightarrow \alpha = 2 \ln 2$$

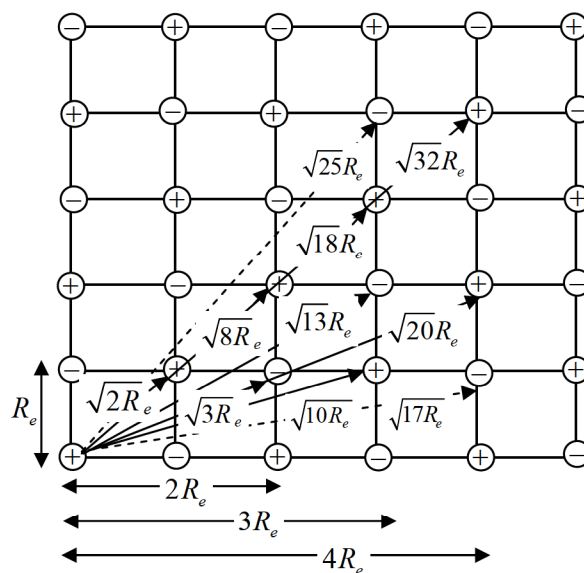
The factor 2 occurs because there are two ions, one to the right and one to the left.

Example: Calculate the Madelung constant for an infinite array of two-dimensional ionic charge distribution (i.e. two dimensional $NaCl$ crystal) with equilibrium ion separation as R_0 .

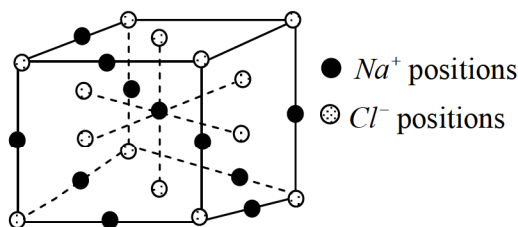
Solution: Consider a two dimensional array of ionic charge (alternate Na^+ and Cl^- ion) distribution as shown as figure. Let us take Na^+ ion (at left bottom corner) as the reference ion and then write various term of the series for alternating ions in term increasing value of R_e , we have

$$A = 4 \left[\left(\frac{1}{1} - \frac{1}{\sqrt{2}} \right) + \left(\frac{1}{\sqrt{4}} - \frac{2}{\sqrt{5}} - \frac{1}{\sqrt{8}} \right) + \left(\frac{1}{3} - \frac{2}{\sqrt{10}} + \frac{2}{\sqrt{13}} - \frac{1}{\sqrt{18}} \right) + \left(-\frac{1}{\sqrt{16}} + \frac{2}{17} - \frac{2}{\sqrt{20}} + \frac{2}{\sqrt{25}} - \frac{1}{\sqrt{32}} \right) + \dots \right]$$

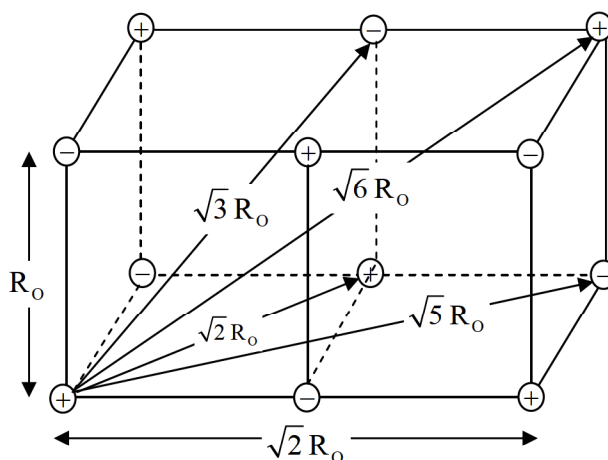
$$= 1.1716 + 0.1635 + 0.0795 + 0.0443 + \dots \approx 1.6135$$



Example: The three dimensional crystal for NaCl is given in figure find the Madelung constant



Solution:



$$\alpha = \frac{6}{1} - \frac{12}{\sqrt{2}} + \frac{8}{\sqrt{3}} - \frac{6}{\sqrt{4}} + \frac{24}{\sqrt{5}} - \frac{24}{\sqrt{6}} + \dots$$

Which will converge to value 1.74

3.3 Bulk Modulus and Compressibility

The bulk modulus is given by $\beta = -V \left(\frac{dP}{dV} \right)$ and compressibility is defined as $\frac{1}{K} = \beta = -V \left(\frac{dP}{dV} \right)$

From first law of thermodynamics $dQ = dU + dW$ at absolute zero $dQ = 0$

So $\frac{dU}{dV} = -P$, $\frac{d^2U}{dV^2} = -\frac{dP}{dV}$ so $\frac{1}{k_0} = V_0 \left(\frac{d^2U}{dV^2} \right)$ where V_0 is volume of crystal at $R = R_0$

For NaCl crystal $R = R_{cl} + R_{Na} = \frac{a}{2}$ and volume per molecule is $= \frac{a^3}{4}$

Therefore $V = \frac{a^3}{4} N = 2NR^3$ so $\frac{dV}{dR} = 6NR^2 \Rightarrow \frac{dR}{dV} = \frac{1}{6NR^2}$

$$\frac{dU}{dV} = \left(\frac{dU}{dR} \right) \cdot \left(\frac{dR}{dV} \right) \Rightarrow \frac{d}{dV} \left(\frac{dU}{dV} \right) = \left(\frac{dU}{dR} \right) \cdot \frac{d}{dV} \left(\frac{dR}{dV} \right) + \left(\frac{dR}{dV} \right) \cdot \frac{d}{dV} \left(\frac{dU}{dR} \right)$$

$$\left(\frac{d^2U}{dV^2} \right) = \left(\frac{dU}{dR} \right) \cdot \left(\frac{d^2R}{dV^2} \right) + \left(\frac{dR}{dV} \right) \frac{d}{dR} \left(\frac{dU}{dR} \right) \cdot \left(\frac{dR}{dV} \right)$$

$$\left(\frac{d^2U}{dV^2}\right) = \left(\frac{dU}{dR}\right) \cdot \left(\frac{d^2R}{dV^2}\right) + \left(\frac{dR}{dV}\right)^2 \cdot \left(\frac{d^2U}{dR^2}\right)$$

At equilibrium separation

$$U = N \left[-\frac{\alpha q^2}{4\pi \epsilon_0 R} + B \exp\left(-\frac{R}{\rho}\right) \right] \text{ For } \left(\frac{dU}{dR}\right) = 0 \text{ and } V = V_0 = 2NR_0^3$$

$$\text{So } \left(\frac{d^2U}{dV^2}\right) = \left(\frac{dR}{dV}\right)^2 \cdot \left(\frac{d^2U}{dR^2}\right) \text{ so } \frac{1}{K_0} = 2NR_0^3 \left(\frac{dR}{dV}\right)^2 \cdot \left(\frac{d^2U}{dR^2}\right)$$

$$\frac{1}{K_0} = 2NR_0^3 \left(\frac{dR}{dV}\right)^2 \cdot \left(\frac{d^2U}{dR^2}\right) \Rightarrow \frac{2R_0^3}{36R_0^4} \left(\frac{d^2U}{dR^2}\right) = \frac{1}{18NR_0} \left(\frac{d^2U}{dR^2}\right) = \frac{\alpha q^2 N}{R_0^3} \left(\frac{R_0}{\rho} - 2\right)$$

$$\frac{1}{K_0} = \beta = \frac{\alpha q^2}{18R_0^4} \left(\frac{R_0}{\rho} - 2\right) \Rightarrow \frac{R_0}{\rho} = \frac{18R_0^4 \beta}{\alpha q^2} + 2.$$

Bond dissociation energy of ionic molecule is given by

$DE = -U_{total} - IP + EA$. where U_{total} potential energy, IP is ionisation potential and EA is electron Affinity.

Example: if $\frac{B}{R^m}$ is repulsive term like $\frac{\rho}{R_0}$ then prove that value of $m = 1 + \frac{72\pi \epsilon_0 R_0^4}{K_0 \alpha q^2}$ and

$$U = -\frac{N\alpha q^2}{4\pi \epsilon_0 R_0} \left(1 - \frac{1}{m}\right)$$

Solution: $U(R) = -N \left[\frac{\alpha q^2}{4\pi \epsilon_0 R} + \frac{B}{R^m} \right] \quad V = V_0 = 2NR_0^3$

Hence, $\frac{dU}{dR} = 0$ at $R = R_0 \Rightarrow R_0^{m-1} = \frac{mB4\pi \epsilon_0}{\alpha q^2} \Rightarrow R_0 = \left(\frac{4\pi \epsilon_0 mB}{\alpha q^2}\right)^{\frac{1}{m-1}}$

$$\frac{1}{K_0} = 2NR_0^3 \left(\frac{dR}{dV}\right)^2 \cdot \left(\frac{d^2U}{dR^2}\right), \quad \frac{dR}{dV} = \frac{1}{6NR^2} \quad \text{Where } \frac{d^2U}{dR^2} = N \left(-\frac{2\alpha q^2}{4\pi \epsilon_0 R_0^3} + \frac{m(m+1)}{R_0^{m+2}} \right)$$

$$\frac{1}{K_0} = 2NR_0^3 \left(\frac{1}{6NR^2}\right)^2 \cdot N \left(\frac{m(m+1)B}{R_0^{m+2}} - \frac{2\alpha q^2}{4\pi \epsilon_0 R_0^3} \right)$$

$$\frac{1}{K_0} = \frac{1}{18R_0^4} \cdot \left(\frac{m(m+1)B}{R_0^{m-2}} - \frac{2\alpha q^2}{4\pi \epsilon_0} \right) \Rightarrow \frac{1}{K_0} = \frac{1}{18R_0^4} \cdot \left(\frac{m(m+1)B\alpha q^2}{mB4\pi \epsilon_0} - \frac{2\alpha q^2}{4\pi \epsilon_0} \right)$$

$$\Rightarrow \frac{1}{K_0} = \frac{\alpha q^2}{72\pi \epsilon_0 R_0^4} [m+1-2] \Rightarrow m = 1 + \frac{72\pi \epsilon_0 R_0^4}{K_0 \alpha q^2}$$

Total energy, $U(R) = -N \left[\frac{\alpha q^2}{4\pi \epsilon_0 R} + \frac{B}{R^m} \right]$ put $R_0^{m-1} = \frac{mB4\pi \epsilon_0}{\alpha q^2}$ and $U = \frac{-\alpha q^2}{4\pi \epsilon_0 R_0} \left[1 - \frac{1}{m} \right]$